

Book I.

Axioms

I.

Magnitudes which are equal to the same are equal to each other.

II.

If equals be added to equals the sums will be equal.

III.

If equals be taken away from equals the remainders will be equal.

IV.

If equals be added to unequals the sums will be unequal.

V.

If equals be taken away from unequals the remainders will be unequal.

VI.

The doubles of the same or equal magnitudes are equal.

VII.

The halves of the same or equal magnitudes are equal.

VIII.

The magnitudes which coincide with one another, or exactly fill the same space, are equal.

IX.

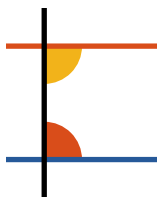
The whole is greater than its part.

X.

Two straight lines cannot include a space.

XI.

All right angles are equal.



XII.

If two straight lines (——— and ———) meet a third straight line (———) so as to make the two interior angles ( and ) on the same side less than two straight angles, these two straight lines will meet if produced on that side on which the angles are less than two right angles.

The twelfth axiom may be expressed in any of the following ways:

1. Two diverging straight lines cannot be both parallel to the same straight line.
2. If a straight line intersect one of the two parallel straight lines it must also intersect the other.
3. Only one straight line can be drawn through a given point, parallel to a given straight line.